

7.3 Classwork: Graphs of Logarithmic Functions — Markscheme

1. [Maximum mark: 10]

(a) Evaluate.

$$(i) \log_4 32 = \log_4(4^{5/2}) = \frac{5}{2} \quad \mathbf{A1}$$

$$(ii) \log_8 4 = \log_8(8^{2/3}) = \frac{2}{3} \quad \mathbf{A1}$$

$$(iii) \log_{1/2} 8: \quad \left(\frac{1}{2}\right)^{-3} = 8, \text{ so } \log_{1/2} 8 = -3 \quad \mathbf{A1}$$

$$(iv) \log_3 \sqrt[3]{9} = \log_3 9^{1/3} = \frac{1}{3} \log_3 9 = \frac{1}{3} \cdot 2 = \frac{2}{3} \quad \mathbf{A1}$$

(b) Let $a > 1$ and $b > 0$. Write each in terms of $\log_a b$.

$$(i) \log_a(a^2 b) = \log_a a^2 + \log_a b = 2 + \log_a b \quad \mathbf{A1}$$

$$(ii) \log_a\left(\frac{b}{a}\right) = \log_a b - \log_a a = \log_a b - 1 \quad \mathbf{A1}$$

$$(iii) \log_a \sqrt{a b^3} = \frac{1}{2} \log_a(a b^3) = \frac{1}{2}(1 + 3 \log_a b) \quad \mathbf{M1}$$

$$= \frac{1}{2} + \frac{3}{2} \log_a b \quad \mathbf{A1}$$

$$(iv) \text{ Change of base: } \log_{a^2} b = \frac{\log_a b}{\log_a a^2} = \frac{\log_a b}{2} \quad \mathbf{M1}$$

$$= \frac{1}{2} \log_a b \quad \mathbf{A1}$$

2. [Maximum mark: 9]

(a) Sketch of $y = 2^x$:

- Correct shape (increasing exponential through $(0, 1)$ and rising to the upper-right)

A1

- y -intercept labelled $(0, 1)$ and horizontal asymptote $y = 0$

A1

(b) Sketch of $y = \log_2 x$ (reflection of $y = 2^x$ across $y = x$):

- Correct shape (passes through $(1, 0)$, increasing, concave down)

A1

- x -intercept labelled $(1, 0)$ and vertical asymptote $x = 0$

A1

(c) Domain and range:

	Domain	Range
$y = 2^x$	$x \in \mathbb{R}$	$y > 0$
$y = \log_2 x$	$x > 0$	$y \in \mathbb{R}$

A1 for each correct cell (**A1 A1 A1 A1**).

(d) Reflection across $y = x$ swaps coordinates. The point on $y = \log_2 x$ corresponding to $(3, 8)$ on $y = 2^x$ is $(8, 3)$.

A1

3. [Maximum mark: 9]

The parent graph is $f(x) = \log_2 x$ with vertical asymptote $x = 0$.

(a) $g(x) = \log_2(x - 2)$ — horizontal shift right by 2.

- Correct sketch (parent shape, shifted +2 in x ; passes through $(3, 0)$, $(4, 1)$, $(6, 2)$)
A1
- Vertical asymptote $x = 2$ **A1**

(b) $h(x) = \log_2 x + 3$ — vertical shift up by 3.

- Correct sketch (parent shape, shifted +3 in y ; passes through $(1, 3)$, $(2, 4)$, $(4, 5)$)
A1
- Vertical asymptote $x = 0$ **A1**

(c) $k(x) = -\log_2 x$ — reflection across the x -axis.

- Correct sketch (parent reflected; passes through $(1, 0)$, $(2, -1)$, $(4, -2)$, $(8, -3)$)
A1
- Vertical asymptote $x = 0$ **A1**

(d) New x -intercept of g : $\log_2(x - 2) = 0 \Rightarrow x - 2 = 1 \Rightarrow x = 3$, so $(3, 0)$. **A1**

(e) $f(a) = \frac{1}{4} \Rightarrow \log_2 a = \frac{1}{4} \Rightarrow a = 2^{1/4} = \sqrt[4]{2}$ **A1**

Point marked on graph at $\left(\sqrt[4]{2}, \frac{1}{4}\right) \approx (1.19, 0.25)$. **A1**

4. [Maximum mark: 3]

(a) $\log_b 27 = 3 \Rightarrow b^3 = 27$ **M1**

$b = 3$ **A1**

(b) The x -intercept of $y = \log_3 x$ is where $\log_3 x = 0$, i.e. $x = 3^0 = 1$. The curve crosses the x -axis at $(1, 0)$. **A1**

5. [Maximum mark: 6]

$y = \log_3(x - h) + k$ has vertical asymptote $x = h$.

(a) Asymptote is $x = 2$, so $h = 2$. **A1**

(b) Substitute $(5, 4)$ into $y = \log_3(x - 2) + k$:

$4 = \log_3(5 - 2) + k = \log_3 3 + k = 1 + k$ **M1**

$k = 3$ **A1**

(c) Sketch of $y = \log_3(x - 2) + 3$:

- Vertical asymptote drawn at $x = 2$ **A1**
- Correct shape (parent $\log_3 x$ shape, shifted right 2 and up 3; passes through $(3, 3)$, $(5, 4)$, $(11, 5)$) **A1**
- Point $(5, 4)$ marked on the curve **A1**